MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

OVERALL SEVERITY: CRITICAL

Case ID	`case01_baseline_vs_case04_t1574_attack`
Generated	`2026-04-09 14:59:33 UTC`
Baseline capture	`case01_baseline`
Attack capture	`case04_t1574_attack`
Baseline path	`/MFF/cases/case01_baseline`
Attack path	`/MFF/cases/case04_t1574_attack`

1. EXECUTIVE SUMMARY

Metric	Value	Metric	Value
New Processes (Attack Only)	15	Critical Risk Processes	6
Gone Processes (Baseline Only)	16	ATT&CK Techniques Observed	3
New Network Connections	94	ATT&CK Tactics Observed	2
Flagged Connections	3	IOCs Extracted	49

ATT&CK; Tactics Observed:

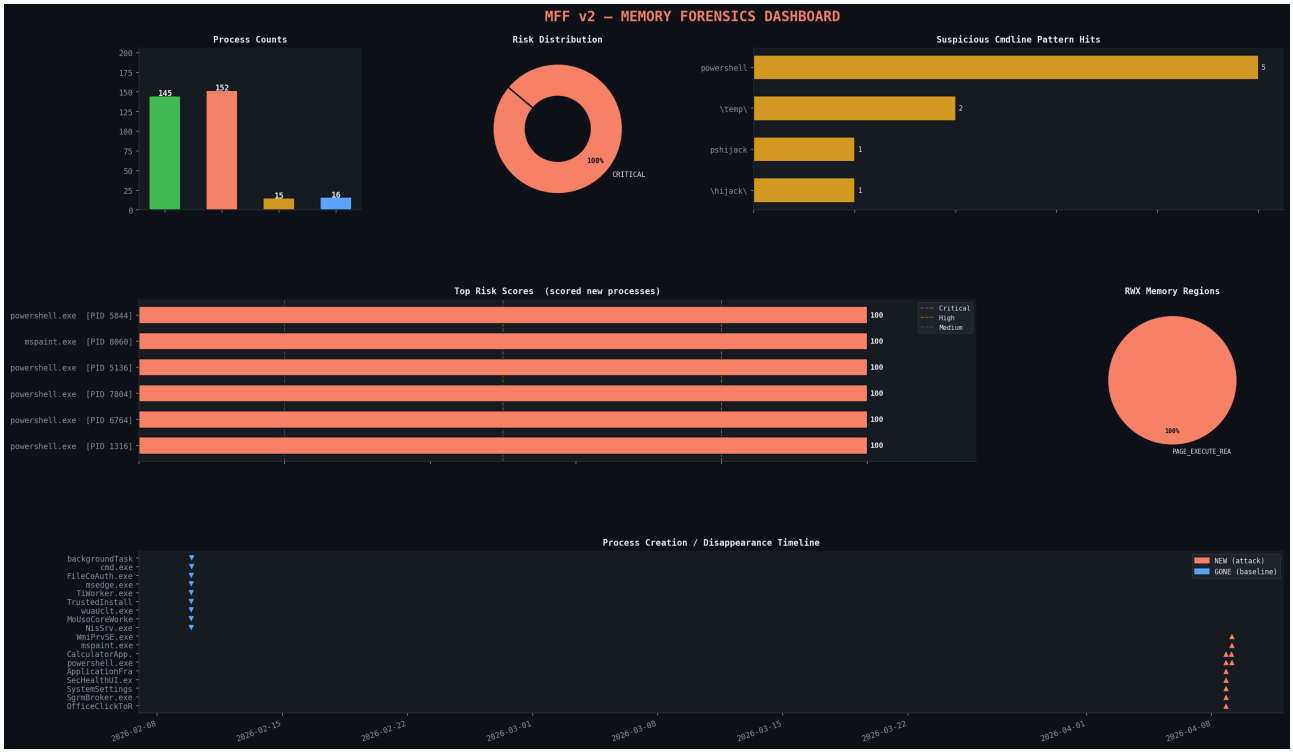
Defense Evasion · Execution

IOC Summary:

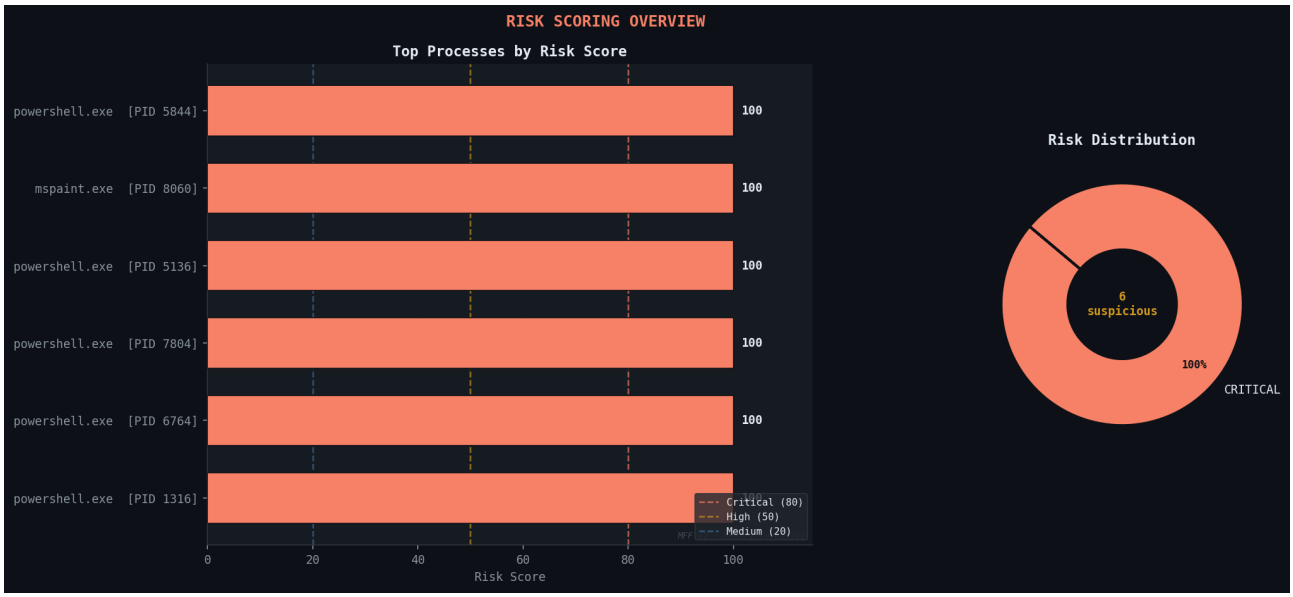
Type	Indicators (top 8)
IPv4	0.0.0.0

2. VISUALIZATIONS

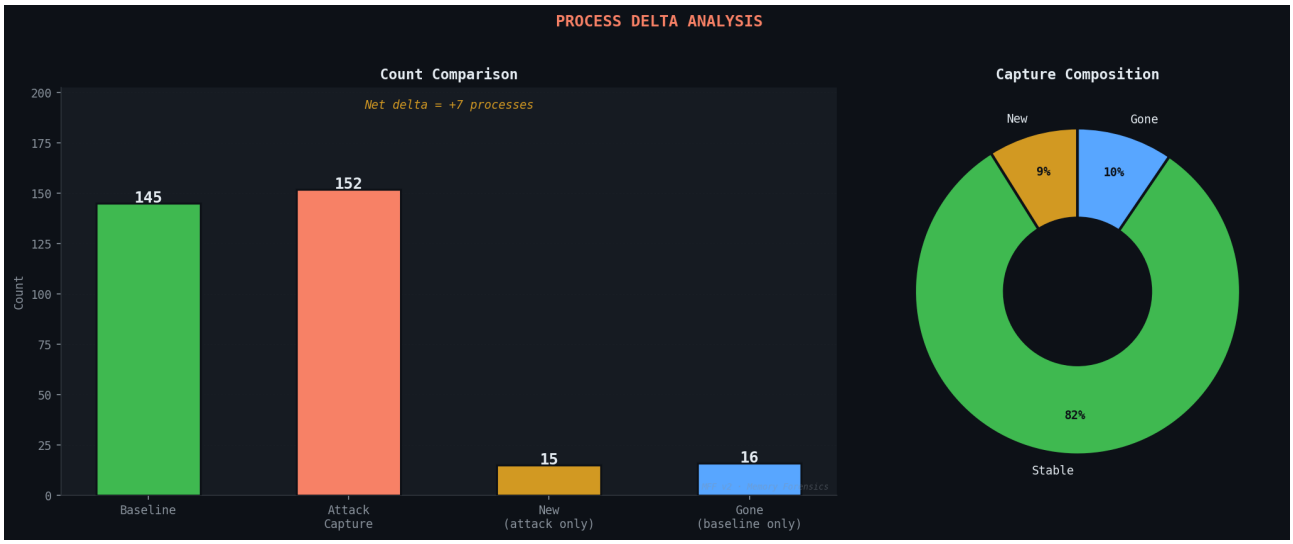
2.1 Full Forensics Dashboard



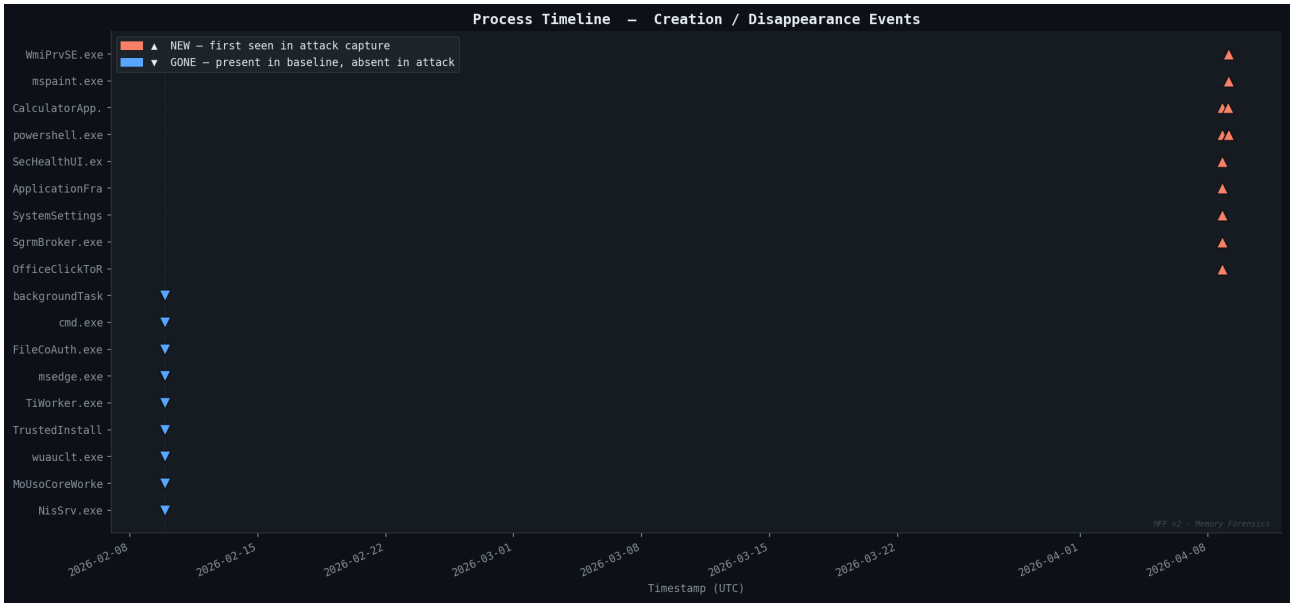
2.2 Risk Scoring Overview



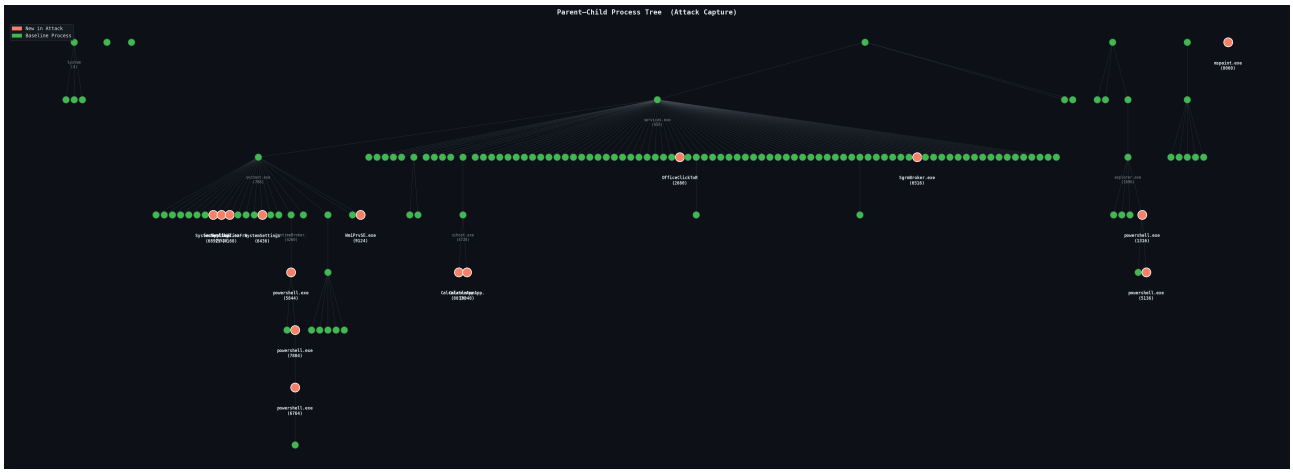
2.3 Process Delta Analysis



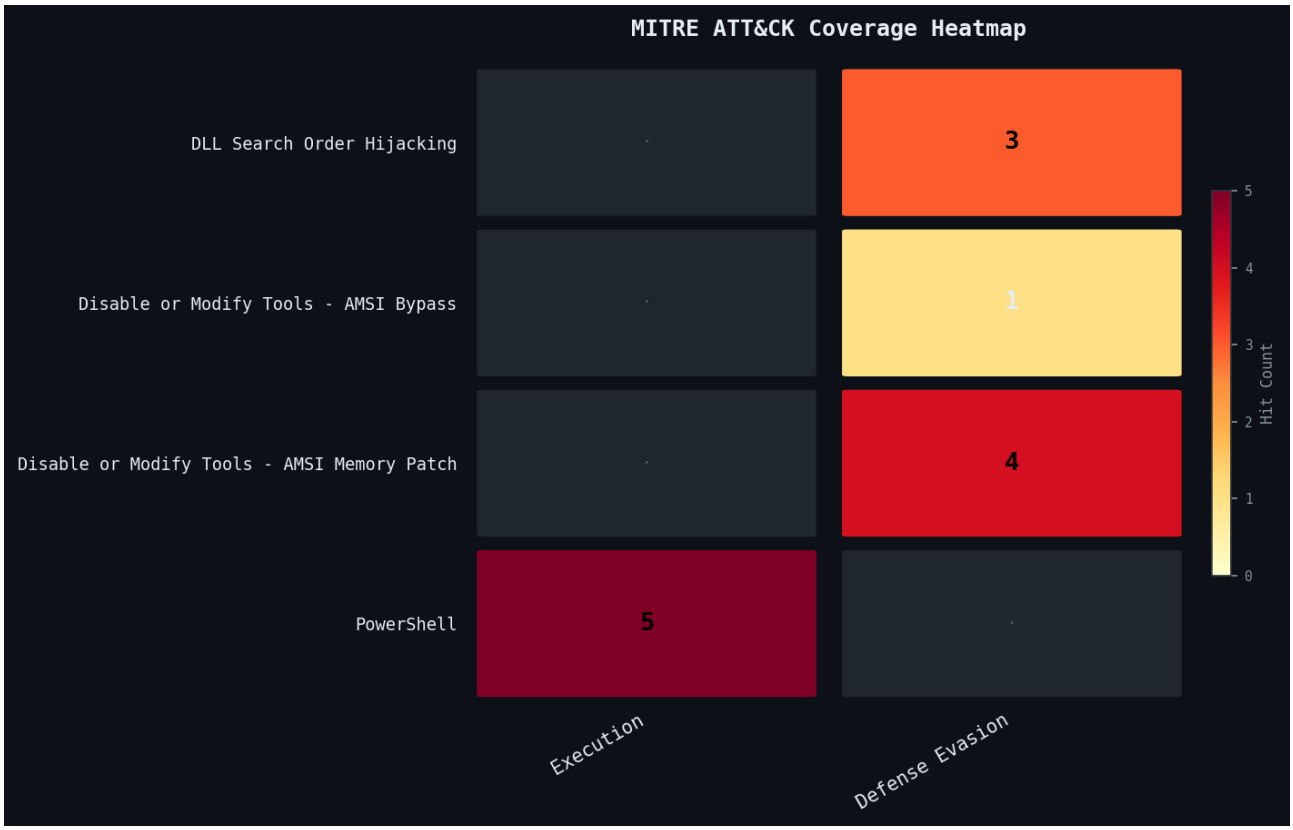
2.4 Process Timeline



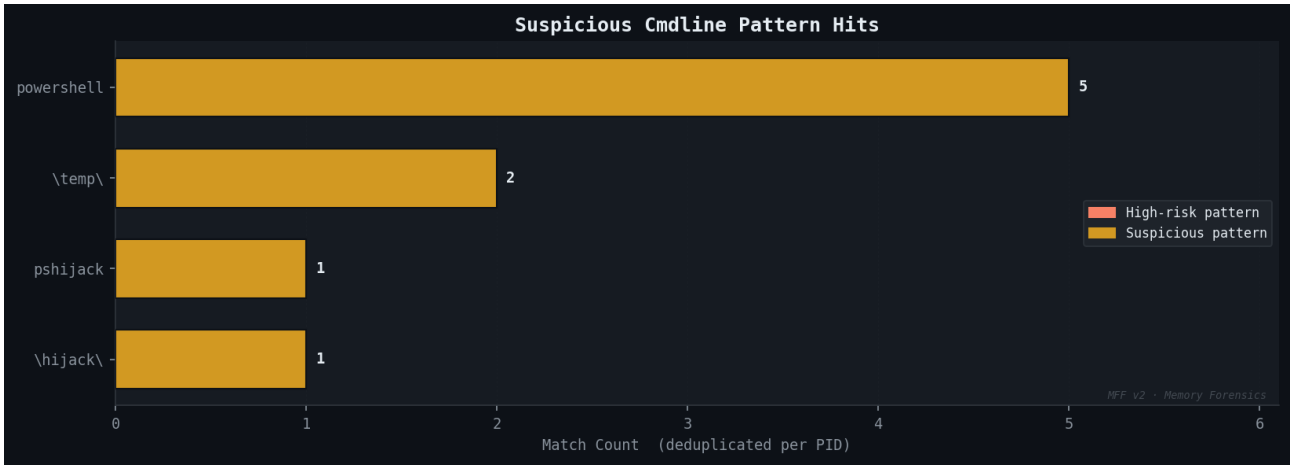
2.5 Parent-Child Process Tree



2.6 MITRE ATT&CK; Heatmap



2.7 Suspicious Cmdline Patterns



2.8 Malfind Memory Protections

MALFIND — SUSPICIOUS MEMORY REGIONS
Memory Protection Types
Region Count
22
0
5
10
15
20
PAGE_EXECUTE_READWRITE
MFF v2 · Memory Forensics
Processes with RWX Regions
powershell.exe
22
0
3
6
9
12
15
18
21
RWX Region Count
MFF v2 · Memory Forensics

3. RISK SCORE TABLE

Process	PID	Score	Level	Indicators		
powershell.exe	5844	100	CRITICAL	T1562.001	3 src	High
mspaint.exe	8060	100	CRITICAL	T1574.001	4 src	Very High
powershell.exe	5136	100	CRITICAL	T1562.001	5 src	Very High
powershell.exe	7804	100	CRITICAL	T1562.001	4 src	Very High
powershell.exe	6764	100	CRITICAL	T1562.001	4 src	Very High
powershell.exe	1316	100	CRITICAL	T1562.001	3 src	High

4. NEW PROCESSES (Attack Only)

ImageFileName	PID	PPID	CreateTime
OfficeClickToR	2680	656	2026-04-08 18:52:03.000000 UTC
SgrmBroker.exe	6516	656	2026-04-08 18:54:12.000000 UTC
SystemSettings	6892	788	2026-04-08 18:54:25.000000 UTC
SecHealthUI.ex	552	788	2026-04-08 18:54:29.000000 UTC
ApplicationFra	4168	788	2026-04-08 18:54:29.000000 UTC
SystemSettings	6436	788	2026-04-08 18:55:10.000000 UTC
powershell.exe	5844	4260	2026-04-08 18:55:14.000000 UTC
CalculatorApp.	8032	4728	2026-04-08 18:55:20.000000 UTC
powershell.exe	1316	1696	2026-04-09 02:18:41.000000 UTC
CalculatorApp.	9048	4728	2026-04-09 02:21:46.000000 UTC
mspaint.exe	8060	4016	2026-04-09 02:34:48.000000 UTC
powershell.exe	7804	5844	2026-04-09 02:39:16.000000 UTC
powershell.exe	6764	7804	2026-04-09 02:39:42.000000 UTC
WmiPrvSE.exe	9124	788	2026-04-09 02:46:13.000000 UTC
powershell.exe	5136	1316	2026-04-09 02:46:42.000000 UTC

5. MITRE ATT&CK; TECHNIQUE MATCHES

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1059.001	PowerShell	Execution	powershell	5844	powershell.exe
T1059.001	PowerShell	Execution	powershell	1316	powershell.exe
T1059.001	PowerShell	Execution	powershell	7804	powershell.exe
T1059.001	PowerShell	Execution	powershell	6764	powershell.exe
T1059.001	PowerShell	Execution	powershell	5136	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Bypass	Defense Evasion	amsi.dll	5136	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	1316	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	5844	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	6764	powershell.exe
T1562.001	Disable or Modify Tools - AMSI Memory Patch	Defense Evasion	amsi.dll	7804	powershell.exe
T1574.001	DLL Search Order Hijacking	Defense Evasion	amsi.dll	5136	powershell.exe
T1574.001	DLL Search Order Hijacking	Defense Evasion	version.dll	8060	mspaint.exe
T1574.001	DLL Search Order Hijacking	Defense Evasion	mspaint.exe	8060	mspaint.exe

6. SUSPICIOUS CMDLINE FINDINGS

PID	Matched Pattern	Args
5844	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
1316	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
7804	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
6764	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
5136	powershell	"C:\\Temp\\pshijack\\powershell.exe"
8060	\temp\	"C:\\temp\\hijack\\mspaint.exe"
5136	\temp\	"C:\\Temp\\pshijack\\powershell.exe"
5136	pshijack	"C:\\Temp\\pshijack\\powershell.exe"
8060	\hijack\	"C:\\temp\\hijack\\mspaint.exe"

7. MALFIND — MEMORY INJECTION ARTEFACTS

Process	PID	Protection
powershell.exe	5844	PAGE_EXECUTE_READWRITE
powershell.exe	5844	PAGE_EXECUTE_READWRITE
powershell.exe	1316	PAGE_EXECUTE_READWRITE
powershell.exe	1316	PAGE_EXECUTE_READWRITE
powershell.exe	1316	PAGE_EXECUTE_READWRITE
powershell.exe	1316	PAGE_EXECUTE_READWRITE
powershell.exe	1316	PAGE_EXECUTE_READWRITE
powershell.exe	7804	PAGE_EXECUTE_READWRITE
powershell.exe	7804	PAGE_EXECUTE_READWRITE

Process	PID	Protection
powershell.exe	7804	PAGE_EXECUTE_READWRITE
powershell.exe	7804	PAGE_EXECUTE_READWRITE
powershell.exe	7804	PAGE_EXECUTE_READWRITE
powershell.exe	6764	PAGE_EXECUTE_READWRITE
powershell.exe	6764	PAGE_EXECUTE_READWRITE
powershell.exe	6764	PAGE_EXECUTE_READWRITE
powershell.exe	6764	PAGE_EXECUTE_READWRITE
powershell.exe	6764	PAGE_EXECUTE_READWRITE
powershell.exe	5136	PAGE_EXECUTE_READWRITE
powershell.exe	5136	PAGE_EXECUTE_READWRITE
powershell.exe	5136	PAGE_EXECUTE_READWRITE
powershell.exe	5136	PAGE_EXECUTE_READWRITE
powershell.exe	5136	PAGE_EXECUTE_READWRITE

7b. DLL ANALYSIS — T1574 / T1562 FINDINGS

9 DLL finding(s) — T1574 DLL Search Order Hijacking + T1562 AMSI Bypass evidence.

PID	Process	DLL	HijackType	Technique	RiskScore	Tactic
5136	powershell.exe	amsi.dll	AMSI_FILE_OUTPUT_DISABLED	T1562.001	90	Defense Evasion
1316	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
5844	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
6764	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
7804	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
5136	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
5136	powershell.exe	amsi.dll	PROTECTED_DLL_USER_DIR	T1574.001	70	Defense Evasion
8060	mspaint.exe	version.dll	PROTECTED_DLL_USER_DIR	T1574.001	70	Defense Evasion
8060	mspaint.exe	mspaint.exe	SYSTEM_EXE_FROM_WRONG_DIR	T1574.001	65	Defense Evasion

[T1562.001] AMSI_FILE_OUTPUT_DISABLED — powershell.exe (PID=5136)

HIGHEST CONFIDENCE T1562.001+T1574.001: 'powershell.exe' (PID=5136) has amsi.dll from 'C:\\Temp\\pshijack\\amsi.dll' (NOT System32). File output='disabled'. Replaced DLL confirmed.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=1316)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=1316) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=5844)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=5844) loads amsi.dll AND has 2 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=6764)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=6764) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=7804)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=7804) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=5136)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=5136) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1574.001] PROTECTED_DLL_USER_DIR — powershell.exe (PID=5136)

```
T1574.001: 'amsi.dll' from staging dir 'C:\\Temp\\pshijack\\amsi.dll' in 'powershell.exe' (PID=5136).
Expected: System32.
```

[T1574.001] PROTECTED_DLL_USER_DIR — mspaint.exe (PID=8060)

```
T1574.001: 'version.dll' from staging dir 'C:\\temp\\hijack\\VERSION.dll' in 'mspaint.exe' (PID=8060).
Expected: System32.
```

8. NETWORK ARTEFACTS

8a. Flagged Connections (suspicious ports)

TreeDepth	Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created	SuspiciousPort	PortMeaning
0	0xba8ee7f531b0	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	904	svchost.exe	2026-04-08 18:51:57.000000 UTC	135	DCOM/RPC
0	0xba8ee7f531b0	TCPv6	::	135	::	0	LISTENING	904	svchost.exe	2026-04-08 18:51:57.000000 UTC	135	DCOM/RPC
0	0xba8ee7f54390	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	904	svchost.exe	2026-04-08 18:51:57.000000 UTC	135	DCOM/RPC

9. IOC DETAIL TABLE

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\svchost.exe	cmdline	48
FilePath	C:\\Windows\\System32\\svchost.exe	cmdline	27
FilePath	C:\\Program	cmdline	20
FilePath	Microsoft\\EdgeWebView\\Application\\146.0.3856.109\\msedgewebview2.exe	cmdline	14
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView	cmdline	7
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.Windows.Search_cw5n1h2txyewy\\LocalState\\EBWebView	cmdline	7
FilePath	C:\\Windows\\System32\\RuntimeBroker.exe	cmdline	6

Type	Value	Source	Count
FilePath	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	4
FilePath	??\\C:\\Windows\\system32\\conhost.exe	cmdline	3
FilePath	C:\\Windows\\system32\\DllHost.exe	cmdline	3
FilePath	C:\\ProgramData\\Microsoft\\Windows	cmdline	2
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.Search_cw5n1h2txyewy\\SearchApp.exe	cmdline	2
FilePath	Microsoft	cmdline	2
FilePath	Windows	cmdline	2
FilePath	WindowsApps\\Microsoft.WindowsCalculator_11.2508.4.0_x64__8wekyb3d8bbwe\\CalculatorApp.exe	cmdline	2
FilePath	system32\\csrss.exe	cmdline	2
FilePath	C:\\Temp\\pshijack\\powershell.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.MicrosoftOfficeHub_8wekyb3d8bbwe\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.Windows.Search_cw5n1h2txyewy\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Windows\\Explorer.EXE	cmdline	1
FilePath	C:\\Windows\\ImmersiveControlPanel\\SystemSettings.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SecurityHealthHost.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SecurityHealthSystray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SystemSettingsBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxService.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxTray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\spoolsv.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.SecHealthUI_cw5n1h2txyewy\\SecHealthUI.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe	cmdline	1

10. METHODOLOGY

Acquisition: VBoxManage debugvm dumpvmcore — cold hypervisor-level acquisition. No kernel module or guest agent. Forensically sound. SHA256 + MD5 hash recorded on host immediately post-acquisition.

Volatility 3: Memory parsed using Volatility 3: pslist, pstree, cmdline, dlllist, malfind, netscan, threads. Exported as CSV (engine) and JSONL (audit trail).

Process diff: Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs are reassigned on every reboot and produce false-positive churn exceeding 90% of all processes.

Risk scoring: Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline pattern +40 · Known-malicious process name +50 · Suspicious DLL load (T1574) +30 · Anomaly name bonus +15. Thresholds: CRITICAL ≥80 HIGH ≥50 MEDIUM ≥20 LOW <20.

Malfind filter: Known-clean JIT processes excluded (browsers, Defender, .NET, OneDrive). Self-match false positives suppressed (e.g. lsass.exe matching itself). Only genuinely suspicious PAGE_EXECUTE_READWRITE regions retained.

DLL analysis: windows.dlllist CSV analysed for T1574 DLL Search Order Hijacking. Protected DLLs (amsi.dll, version.dll, cryptbase.dll etc.) checked for loads from non-System32 paths. Findings merged into ATT&CK; tagged output.

ATT&CK; mapping: 30+ rule-based signatures across 8 MITRE ATT&CK; tactics. Keyword matching on cmdline args, process names, DLL paths, and network artefacts. T1574-specific patterns include: copy-item, amsi.dll, invoke-atomictest.

IOC extraction: Regex-based extraction: IPv4 (excluding RFC1918 private ranges), domains, URLs, MD5/SHA1/SHA256 hashes, Windows file paths.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project — University of Roehampton 2026. Tool chain: VBoxManage → Volatility 3 → MFF v2 → HTML + PDF reports. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, export_alert, report_generator.

Risk scoring: Transparent additive model — no ML. Each evidence category contributes once. Weights calibrated to specificity (ACPO P2). CRITICAL>=80, HIGH>=55, MEDIUM>=25.

Standards: ACPO Good Practice Guide for Digital Evidence (Principles 1-4). NIST SP 800-86 Integrating Forensic Techniques into Incident Response. ISO/IEC 27037:2012 Guidelines for Identification, Collection and Preservation. College of Policing Digital Forensics Guidance. CPS Disclosure Manual.